Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/27/2007
PI Name:	Dunn-Rankin, Derek Ph.D.		
Project Title:	Applications of Electric Fields in Microgravity Combustion Science		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ddunnran@uci.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-8745
Organization Name:	University of California - Irvine		
PI Address 1:	Department of Mechanical & Aerospace Engineering		
PI Address 2:	4200 Engineering Gateway Bldg, EG3224		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-3975	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: grant reinstatement resulted in new grant--see task with same title appended with NNX07AB55A and period of performance of 10/01/2006-9/30/2009, per D. Stocker/GRC (10/07). NOTE: grant reinstated 8/2006; end date now 9/30/2007, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Weinberg, Felix (Imperial College of Science, Technology, and Medicine)		
Grant/Contract No.:	NNC04GA28G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Objectives: This project explores the use of large electric fields for a wide variety of space exploration applications, primarily those involving ion-driven winds. The specific objectives of the study that we concentrated on this year include: 1) Experimentally determine the extent to which electric fields can simulate a variable gravity condition (e.g., Mars gravity, moon gravity, zero gravity) for flames in an Earth-based laboratory so as to study phenomena that will be important in these space and planetary environments. 2) Examine the fundamentals of ion wind interactions with flames and their effects on flame behavior in order to understand the practicality of controlling flames electrically. 3) Determine active control methods to attain and maintain effective g-loadings on flames by ion wind actuation and quantify the extent of this action; such a method can be demonstrated by removing g-jitter from combustion experiments in parabolic flights. 4) Construct a dual-electrostatic burner (i.e., no moving parts) in which a flame is sustained and oxygenated by an air swirl generated from an ion wind and liquid fuel is delivered via wick or electrostatic atomization. 5) Optimize ion wind generation for space-based combustion applications and investigate concepts for maximizing flow. 6) Use ion winds to enhance the performance of electrostatic precipitators. 7) Evaluate the potential for hand-held electrosprays to extinguish fires in a variety of gravity-levels. Task Description: In order to achieve the above objectives, there are two principal thrusts to the research. One focuses on the electrically controlled flame system, where experiments are conducted in an Earth based laboratory enabling a wide range of non-intrusive diagnostics on flames in various g-loading conditions including that equivalent to microgravity. Diagnostics such as schlieren, shadowgraph, deflection imaging, are applied to visualize a burning methane-air flame under the influence of electrically induced microbuoyancy. Numerical simulations predict the forces involved and allow new designs for enhanced flame control. For the ion wind driven burner and extinguisher systems, it is necessary to understand the limitations of ion winds. For example, it is to be determined whether or not the airflow can be accelerated with the addition of a nozzle and/or if the blowers are staged. From the results, an ionic air swirler is to be designed where a vortex is formed within a tube containing air at STP with adequate strength to hold a flame. The vortical flow can also be applied to a fan-less electrostatic precipitator capable of enhanced particle removal efficiency by combining electrostatic and centrifugal forces on the particles entrained into the system. Task Significance: Understanding electric field interactions with flames is central to combustion control, a technology that not has application in advanced propulsion and power systems, but also toward fire safety in space and on other planets. Appropriately applied fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our facility provides a unique ability to make simultaneous measurements of temperature, species concentration, flow patterns, and ion currents. One goal central to NASA’s mission is to improve spacecraft and remote habitat fire safety for travel and off-world activities. The ion wind and electrospray work is expected to reveal new and improved ways to prevent, detect, and extinguish fires. In addition, advances in small-scale combustion systems capable of providing compact power can be developed without moving parts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The electric control of flames can be used to modfiy combustion near the limits of flammability. At these limit conditions, the flame is very sensitive to small changes in local environment, and forces acting on flame ions produce flows directly in the reaction zone with little time delay. Several of the applications we explore have potential uses in earth gravity as well as microgravity. For example, the application of liquid dispersions of biologically active material for agriculture can reduce the use of chemical toxins. Understanding fundamental electrical properties of flames will also have contributions to earth-based combustion though the forces from electric fields will be less dramatic in the presence of natural convection. One innovative technology that has come out of the research is a microscale ion-wind generator that may have applications for local cooling of electronics or for gaseous microfluidics.
	Taking the specific objectives in turn, the following progress has been achieved over the first year of the project -- • Experimentally determine the extent to which electric fields can simulate a variable gravity condition – We have come to some closure on this question by studying the coupling between ion wind and the small flames that produce them. These results fit well with those from the following section and so they are combined there. • Examine the fundamentals of ion wind interactions with flames – Through the culmination of a detailed experimental and computational effort, most of the key elements of ion wind interactions with small diffusion flames have been elucidated. The details of these findings appear in a dissertation (Papac, 2005). The computational model used the COMSOL multiphysics solver to simulate the coupling between the electrostatics, electrokinetics, and fluid flow. We simplified the problem by assuming a small spherical cap flame as seen in the experiment. The experiment consists of a small capillary feeding methane gas between two wire gauze planes. The wire mesh near the flame is held at a high positive voltage relative the opposite plane. The ground plane is held at slightly higher voltage than true ground in order to measure the ion current. A small collection probe translates along the ground plane to compare the ion current distribution along the ground plane and the analytical solution describing that current. Remarkably, the ion current distribution follows very closely the classical Warburg distribution that is found for ion currents in DC corona discharges. The Warburg distribution occurs as a result of strong space charge effects in the vicinity of the ion source, implying that the small diffusion flames act essentially as a relatively high density source of ions (though the ion number concentration is significantly smaller than is found in coronas). Perhaps most importantly, the computational results show the limitations on using electric fields to simulate complete microgravity conditions. The experiment also allowed a comparison between luminous images and a schlieren images of the electrically actuated flame at different potentials. The research also included a full numerical simulation that includes fluid flow, buoyancy, and ion generation. The total ion generation is set so that it matches the experimental value for the balanced condition. From the computation, we found that the hot rising gas above the flame coupled with the rising gas that was initially drawn downward by the ion wind creating a shear layer a few flame diameters from the capillary. This shear layer location matches precisely that found in our earlier work measuring the temperature profile near the flame. Because the ion transport profile is dominated near the flame by self-repulsion (i.e., space charge) different electrode shapes will not have a significant effect on this behavior, suggesting that there is an intrinsic limit to the spatial extent of microgravity that can be simulated using electric fields. Nevertheless, along the axis of the system, it is possible to produce a high degree of buoyancy cancellation (i.e., microgravity conditions) that is protected by a stagnant sheath over approximately 3 flame diameters.

Task Progress:	• Determine active control methods – One of the principal findings of the prior item was the relationship between the ion production and the changes to the flame that are driven by the ion wind that results. This realization provided us with the view of a flame as an active electrical element. That is, the resistance to current flow through the flame can be considered a non-linear function of the voltage. The VI curve for our small diffusion flame shows a quadratic relationship between ion current and voltage in the subsaturated range, an essentially constant ion current during a saturation plateau, and then a secondary increase in current beyond the plateau. Originally we thought this secondary increase was caused by secondary ionization, but we now suppose it to result from changes in the flame behavior that produces higher ion production. Taking the ratio of the voltage by current, we can simulate the flame as an electric element. Furthermore, by exposing the flame circuit to step voltage changes, the effective resistance and capacitance can be determined precisely. These values can then be implemented into a feedback control loop to drive the ion current through the system along any desired trajectory. The demonstration of these findings were presented in the Spring Meeting of the Western States Section/The Combustion Institute (Borgatelli and Dunn-Rankin, 2006) and is the core of Borgatelli's thesis. • Construct a dual-electrostatic burner – The major progress in the burner design was presented at the recent Combustion Symposium (Weinberg, et al., 2006). In this work, we combined a compact ion wind generator (discussed more in the next item) with a wick fed by a liquid fuel. Following assessment of the various air blower designs for the burner, a single stage of 25mm diameter was chosen (though the others may prove useful for different requirements). At its length of just over 12.5 mm this is less than 7ml in volume and, with an electrical input of around 0.5W, delivers approximately 1 l/s air, corresponding to a calculated maximum thermal output of 3 - 3.5kW for stoichiometric hydrocarbon mixtures. Various gaseous and liquid fuels are injected immediately above the earthed ring electrode through surgical steel capillaries. Short premixed flames are stabilized using a combination of wicks of ceramic wool which, depending on the air flow rate, can also act as porous plug burners and bluff body flame stabilizers. Flames for a range of flow rates of CnH2n+2 fuels with n = 1, 3, 6, 10 as well as with methylated spirit were stabilized. A study of minimum air flows required to avoid soot formation leads to the conclusion that, for a given wick area, the largest thermal power outputs are achieved with those fuels that are least prone to sooting. In microgravity, capillarity and/or their vapour pressure acting in a container would need to replace gravity feed for liquid fuels. • Optimize ion wind generation – Substantial progress has been made on understanding the performance of needle-ring ion wind generators. In addition to helping improve electrostatic burner described under the prior bullet, the combined experimental and computational study of corona ion wind generation formed the basis for the filing of a recent patent application (see below). Most of the details for the work in this category of the project appear in a dissertation (Rickard, 2005 – see below). The computations were developed in the COMSOL Multiphysics software package. The comparison between the predicted velocity profile and the measured profile. There are some discrepancies near the axis, but the model accurately predicts the velocity performance of these devices, and shows that they produce a strong axial jet flow that gradually mixes into the remaining air. • Use ion winds to enhance the performance of electrostatic precipitators – Based on our prior study, the contribution of ion winds to the transport of charged particles is small and is unlikely to be significant. Consequently, we focus our efforts on other aspects of the project. • Evaluate the potential for hand-held electrosprays to extinguish fires – We have not yet examined further this feature of the project. In preparation, however, we are continuing to examine electrospray performance in general; we continue to explore electrospray potential for distributing biologically active materials in thin films. In addition, we have tested electrosprays of water to humidify the air being drawn into a small engine. These represent cross-over applications for the generation of fine sprays in microgravity. SUPPLEMENTAL INFORMATION--OCTOBER 2007 Grant reinstatement resulted in new grant--see task with same title appended with NNX07AB55A and period of performance of 10/01/2006-9/30/2009, per D. Stocker/GRC (10/07).

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Patents	US Patent application. US Patent application was filed on 5/31/2006. May-2006 Dunn-Rankin D et al. "High-Velocity, Multistage, Nozzled, Ion Driven Wind Generator And Method of Operation of the Same Adaptable to Mesoscale Realization."